

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-2

Step 1 — Create Database

```
[mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.00 sec)

[mysql> USE bookflow_db;
Database changed
```

Step 2 — Create Books Table (DDL + Constraints)

```
mysql> CREATE TABLE Books (
->     book_id INT PRIMARY KEY,
->     title VARCHAR(100) NOT NULL,
->     isbn VARCHAR(20) UNIQUE,
->     published_year INT CHECK (published_year < 2027)
[    -> );
Query OK, 0 rows affected (0.01 sec)
```

Step 3 & 4 — Insert Books and Display

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
-> (1, 'The Great Gatsby', '9780743273565', 1925),
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
-> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn                | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565      | 1925          |
| 2       | To Kill a Mockingbird | 9780061120084      | 1960          |
| 3       | 1984                 | 9780451524935      | 1949          |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Step 5, 6 & 7 — Members Table: Create, Insert, Display

```
mysql> CREATE TABLE Members (  
->     member_id INT PRIMARY KEY,  
->     full_name VARCHAR(100),  
->     email VARCHAR(100) UNIQUE  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> INSERT INTO Members (member_id, full_name, email) VALUES  
-> (101, 'John Smith', 'john.smith@email.com'),  
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),  
-> (103, 'Michael Brown', 'michael.brown@email.com');  
Query OK, 3 rows affected (0.02 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> SELECT * FROM Members;  
+-----+-----+-----+  
| member_id | full_name   | email                     |  
+-----+-----+-----+  
|         101 | John Smith  | john.smith@email.com     |  
|         102 | Emma Wilson | emma.wilson@email.com    |  
|         103 | Michael Brown | michael.brown@email.com  |  
+-----+-----+-----+  
3 rows in set (0.00 sec)
```

Step 8 — Create Loans Table (FOREIGN KEYS)

```
mysql> CREATE TABLE Loans (  
->     loan_id INT PRIMARY KEY,  
->     member_id INT,  
->     book_id INT,  
->     loan_date DATE,  
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),  
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

Step 9 & 10 — Insert 10 Loans and Display

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
-> (1, 101, 1, '2025-01-05'),  
-> (2, 102, 2, '2025-01-08'),  
-> (3, 103, 3, '2025-01-10'),  
-> (4, 101, 2, '2025-02-01'),  
-> (5, 102, 1, '2025-02-05'),  
-> (6, 103, 2, '2025-02-12'),  
-> (7, 101, 3, '2025-03-01'),  
-> (8, 102, 3, '2025-03-07'),  
-> (9, 103, 1, '2025-03-15'),  
-> (10, 101, 1, '2025-04-01');  
Query OK, 10 rows affected (0.00 sec)  
Records: 10 Duplicates: 0 Warnings: 0  
  
mysql> SELECT * FROM Loans;  
+-----+-----+-----+-----+  
| loan_id | member_id | book_id | loan_date |  
+-----+-----+-----+-----+  
|      1 |      101 |      1 | 2025-01-05 |  
|      2 |      102 |      2 | 2025-01-08 |  
|      3 |      103 |      3 | 2025-01-10 |  
|      4 |      101 |      2 | 2025-02-01 |  
|      5 |      102 |      1 | 2025-02-05 |  
|      6 |      103 |      2 | 2025-02-12 |  
|      7 |      101 |      3 | 2025-03-01 |  
|      8 |      102 |      3 | 2025-03-07 |  
|      9 |      103 |      1 | 2025-03-15 |  
|     10 |      101 |      1 | 2025-04-01 |  
+-----+-----+-----+-----+  
10 rows in set (0.00 sec)
```

Step 11 — JOIN Query (Who Borrowed What)

```
mysql> SELECT m.full_name AS Member_Name,  
->      b.title AS Book_Title  
-> FROM Loans l  
-> INNER JOIN Members m  
->      ON l.member_id = m.member_id  
-> INNER JOIN Books b  
->      ON l.book_id = b.book_id;  
+-----+-----+  
| Member_Name | Book_Title |  
+-----+-----+  
| John Smith  | The Great Gatsby |  
| John Smith  | To Kill a Mockingbird |  
| John Smith  | 1984 |  
| John Smith  | The Great Gatsby |  
| Emma Wilson | To Kill a Mockingbird |  
| Emma Wilson | The Great Gatsby |  
| Emma Wilson | 1984 |  
| Michael Brown | 1984 |  
| Michael Brown | To Kill a Mockingbird |  
| Michael Brown | The Great Gatsby |  
+-----+-----+  
10 rows in set (0.00 sec)
```

Step 12 — GROUP BY Query (Books per Year)

```
mysql> SELECT published_year,  
->      COUNT(book_id) AS Total_Books  
-> FROM Books  
-> GROUP BY published_year  
-> ORDER BY published_year;  
+-----+-----+  
| published_year | Total_Books |  
+-----+-----+  
| 1925 | 1 |  
| 1949 | 1 |  
| 1960 | 1 |  
+-----+-----+  
3 rows in set (0.00 sec)
```

Step 13–16 — Donation via TRANSACTION (Atomicity)

```
mysql> CREATE TABLE Donation_History (  
->      donation_id INT PRIMARY KEY,  
->      book_id INT,  
->      donor_name VARCHAR(100),  
->      donation_date DATE,  
->      FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
[mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> INSERT INTO Books (book_id, title, isbn, published_year)  
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO Donation_History  
-> (donation_id, book_id, donor_name, donation_date)  
-> VALUES (1, 4, 'Raj Kumar', CURDATE());  
Query OK, 1 row affected (0.00 sec)  
  
[mysql> COMMIT;  
Query OK, 0 rows affected (0.00 sec)
```

Step 17 & 18 — Index on ISBN + Fast Search

```
mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books
-> WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
| 3       | 1984  | 9780451524935 | 1949           |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Final State of All Tables

```
[mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| Books                  |
| Donation_History       |
| Loans                  |
| Members                |
+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title          | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984           | 9780451524935 | 1949           |
| 4       | Animal Farm     | 9780451526342 | 1945           |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name      | email          |
+-----+-----+-----+
| 101       | John Smith     | john.smith@email.com |
| 102       | Emma Wilson    | emma.wilson@email.com |
| 103       | Michael Brown  | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1       | 101       | 1       | 2025-01-05 |
| 2       | 102       | 2       | 2025-01-08 |
| 3       | 103       | 3       | 2025-01-10 |
| 4       | 101       | 2       | 2025-02-01 |
| 5       | 102       | 1       | 2025-02-05 |
| 6       | 103       | 2       | 2025-02-12 |
| 7       | 101       | 3       | 2025-03-01 |
| 8       | 102       | 3       | 2025-03-07 |
| 9       | 103       | 1       | 2025-03-15 |
| 10      | 101       | 1       | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

[mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1           | 4       | Raj Kumar  | 2026-07-29    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```